

Differentiating with exponentials and logarithms 0.3.9

$$\frac{d}{dx} e^x = e^x \quad \text{from last video}$$

What about a^x ? Key is that $a = e^{\ln(a)}$

so that $a^x = e^{x \ln(a)}$.

$$\text{Then } \frac{d}{dx} a^x = \frac{d}{dx} e^{x \ln(a)} = e^{x \ln(a)} \cdot \ln(a) = a^x \ln(a)$$

$$\boxed{\frac{d}{dx} a^x = a^x \ln(a)}$$

Example: Compute $\frac{d}{dx} 3^{\sin(x)}$.

$$\frac{d}{dx} 3^{\sin(x)} = 3^{\sin(x)} \ln(3) \cdot \cos(x) = \ln(3) \cos(x) \cdot 3^{\sin(x)}$$

Recall the Inverse Function Theorem:

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

$$\frac{d}{dx} \log_a(x) = \frac{1}{a^{\log_a(x)} \cdot \ln(a)} = \frac{1}{x \cdot \ln(a)}$$

$$\boxed{\frac{d}{dx} \log_a(x) = \frac{1}{\ln(a) x}}$$

$$\boxed{\frac{d}{dx} \ln(x) = \frac{1}{x}}$$

Example: Compute $\frac{d}{dx} \ln(x^3 + 3x - 4)$.

$$\frac{d}{dx} \ln(x^3 + 3x - 4) = \frac{1}{x^3 + 3x - 4} \cdot (3x^2 + 3) = \frac{3x^2 + 3}{x^3 + 3x - 4}$$

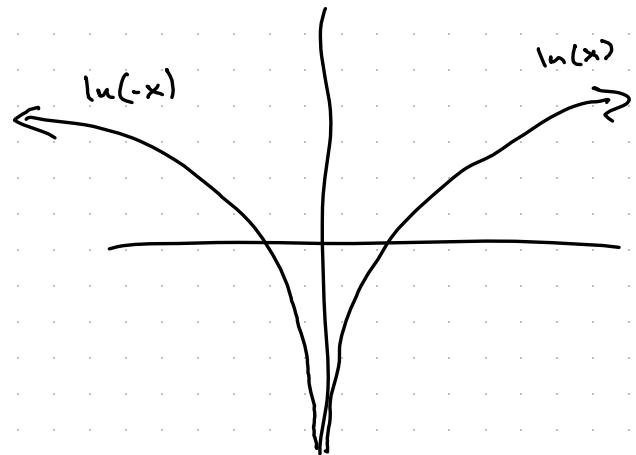
Idea: let's consider $\ln(|x|)$

$$\frac{d}{dx} \ln(-x) = \frac{1}{-x} \cdot -1 = \frac{1}{x}$$

$$\boxed{\frac{d}{dx} \ln|x| = \frac{1}{x}}$$

$$\frac{d}{dx} \ln|x| = \frac{1}{|x|} \cdot \frac{d}{dx} |x| = \dots ?$$

Exercise!



together we have
a graph for $\ln(|x|)$

Domain: $(-\infty, 0) \cup (0, \infty)$